

Taiwan Explorers Final Report (M2)

GROUP 18 FLOWERS

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GITHUB REPOSITORY LINK

<https://github.com/CMPT-276-SUMMER-2025/final-project-18-flowers>

DEPLOYED WEBSITE LINK

<https://taiwanexplorers.onrender.com/>

Project Delivery

1. Project Success

Provide analysis of the project's success in meeting the user needs and solving the problems you've stated. (include feedback from in-person testing session)

Before and during development, our team identified three key problems we aimed to address:

- A lack of high-quality, Taiwan-focused travel guides tailored for Western or English-speaking audiences.
- Trip planning for Taiwan is often time-consuming and overwhelming for users.
- Reliable, clearly organized information about Taiwan's attractions that is difficult to find.

To tackle these issues, we created a travel website dedicated to Taiwan, featuring destination pages, interest-based attraction filters, and an itinerary generator. Our primary goal by the deadline was to implement:

- A dedicated city/county destinations page
- A categorized attractions page (sorted by interest)
- An itinerary generator
- A well-designed, fully functional web app with core features like search and responsive layout

Given the lack of Taiwan travel websites, our web application was created to make planning a trip to Taiwan easier and more accessible for English audiences. All content is presented with the presence of English, and we carefully filtered and cleaned up the place data from the Google Maps API to ensure that every location and description shown is relevant.

To make trip planning as quick and efficient as possible, we built an itinerary generator that walks users through a few key input fields. Once submitted, it automatically generates a travel plan that includes a list of destinations and a visual route map. The map not only shows the travel path but also estimates travel times using different modes of transportation like driving, public transit, cycling, or walking.

This gives users a clear picture of how their overall trip will look and give a gauge of the total time.

We also focused heavily on design usability to make sure the experience feels smooth and intuitive from start to finish. The layout follows established design principles, using visual hierarchies, and balanced margins/padding to reduce clutter. Most pages are fully responsive and work well on both desktop and mobile devices. However, the itinerary generator is still in development for smaller screens and will receive further mobile optimizations in future updates.

Overall, we believe we achieved the majority of our goals. All core pages were completed and matched closely with our Figma MFP. Due to time constraints, however, we had to omit some components that weren't part of our core features, such as the "Plan a trip" section on the home page or the "money slider input" for the itinerary generator.

Feedback from Testing and Peer Review

During our in-person peer review session, feedback was largely positive. Most users rated their experience as "good" or "perfect," especially praising the user-friendly interface. The booking component initially had a minor bug related to the external query link, but we addressed and resolved it promptly.

Users particularly appreciated the website's overall UI and clean, accessible layout. A few areas for improvement were also identified:

- Some users found font sizes too small or character spacing too tight on certain pages.
- The itinerary generator layout and response content could be improved for better clarity and usefulness.
- The generator currently only supports USD inputs, limiting accessibility for international users.
- One requested feature, saving or downloading itinerary results, was not implemented. We agree this is a valuable addition and have noted it down as a future feature we should implement.

2. Analysis of SDLC Model

Provide analysis of the SDLC model and how it was used in the project.

Back in milestone 1, our group decided to use the Agile Kanban SDLC model. In the end, we decided to stick with this model, as it provided many critical benefits. During the planning phase of this project, our group realized that each member had different strengths and weaknesses that we had to plan for accordingly. Due to the roles-based nature of Agile, our group used this model to minimize the impact of our shortcomings and fully exploit our strengths. For example, we divided the task of implementing the API 1 and 2 features into two teams: Alex and Ryan worked on API 1 features, while Tian and Ikkyu worked on API 2 features. Instead of trying to get four people to work on the same feature, we used a divide and conquer strategy by assigning roles to our group members to improve efficiency. The Agile Kanban model is better suited for adapting to change, which was a necessity. Our group used standups and retrospective meetings to discuss our progress to ensure that members were on track.

Additionally, we would discuss whether any changes were needed. The current iteration of our website differs from the original plan that we envisioned in many ways. Owing to the flexibility of this model, we were comfortable with making these changes as long as the whole group was in agreement. An example of a change would be the layout of the itinerary generator page, as we decided that the old layout did not meet our quality standards. Another example is how our group decided to replace the street view feature with the route data feature, as we felt that it was unbeneficial to users. Finally, we used GitHub Issues' built-in Kanban board to create cards for every task. This aspect of Agile Kanban kept every group member focused on what was important. While working on this project, we adhered to the Agile manifesto's message of "responding to change over following a plan," as our group made slight tweaks to the project as needed instead of following an exact plan. These actions allowed us to advance to the next stage instead of being stuck. Agile Kanban was an extraordinarily effective SDLC model that provided us with flexibility, responsibility, and collaboration. As such, we followed this SDLC model to a tee.

3. API Features

API 1: Google Maps API

Link: [Google Maps Platform Documentation](#) | [Google for Developers](#)

Feature 1: Integrate Google's Vast Database on Places

Display location and information data about various attractions, accommodations, and more using Google Maps API's Places API.

- **Benefits:**

- Users will be able to see what places look like and gain a first impression of the place to consider for their trip.
- Users will be able to find accommodations near places to find hotels, restaurants, or stores.
- Users will be able to get information such as a description, honest reviews, and cost to make wise decisions and create fulfilling personal plans.

Detailed Description:

We integrated Google's Places API to fetch data such as photos, place IDs, reviews, editorial summaries, amenities, place types, and more. This data allowed us to generate detailed pages for destinations dynamically. For example, users can navigate to a dedicated page for Taipei City from the destinations page. On each destination page, we showcase nine of the most popular attractions in the area along with highly rated hotels and restaurants. The same data is also used on the attractions page, where we display attraction photos, weekly opening hours, editorial summaries, and nearby amenities. All of these pages, including those accessed through the interests page, are dynamically generated. As long as we provide an attraction name, the page can be automatically created and populated with all relevant data.

Feature 2: Interactive Dynamic Maps and Static Maps

Display interactive maps of various locations or routes that include markers with popups, custom styling, and customizability of data displayed on the map.

- **Benefits:**

- Users will be able to click on markers to get helpful information immediately through content cards.
- Users will get a general sense of every attraction's location through static maps.
- Users will be able to visually explore different locations and routes engagingly by clicking buttons or interacting with the map through zoom or drag.

Detailed Description:

In our application, we included an interactive map that highlights various destinations across Taiwan, such as Taipei City, Hualien City, and Yilan City. Each destination is marked with an interactive point that displays an information card when clicked, helping users quickly understand key details about the location. We also used static maps on the individual attraction pages, such as for the Taipei 101 Observatory, to

give users a basic sense of each attraction's location. Additionally, we implemented a dynamic map component in the itinerary generator. This map uses coordinates generated during the itinerary process to place markers at relevant locations. Users can zoom, pan across the map, and use interactive buttons to filter the itinerary view by travel mode, which displays different route options.

Feature 3: Return Route Data and Display It

Calculate and display routes between a series of markers and return specific directions for transportation such as transit, driving, walking, and more.

- **Benefits:**
 - Users will be able to see visually how long it'll take to traverse between various locations and fit it into their travel plan.
 - Users will be able to compare the length of a route based on the mode of transportation.
 - Users will be able to gauge the total time.

Detailed Description:

In our itinerary generator map, we added routes using multiple transportation modes to help users efficiently plan their travel and visualize the locations of the suggested attractions. The process works as follows: the user provides destination names, such as Taipei City or Hualien City, and for each destination, the system generates a list of attractions. Using this data, the itinerary map displays the attractions with numbered markers, based on their order in the itinerary, and dynamically creates routes between them according to the selected mode of transportation. In the bottom corner of the map, we use the Directions Service to calculate and display the duration of each route based on the chosen travel mode. This feature gives users a clear, interactive visualization of their trip layout, helping them better understand the total travel time and plan accordingly.

API 2: Google Gemini API

Link: <https://ai.google.dev/gemini-api/docs>

Feature 1: Itinerary Generator

AI trip planner with Google Services Integrations by utilizing Google Gemini API and Google Maps API. Users can personalize their itinerary with factors such as time, cost, location, and interests.

- **Benefits:**

- Users will be able to receive a personalized travel plan based on their schedule, budget, and preferences.
- Users will be able to save time by having an AI automatically generate optimized routes based on the users' input.
- Users will be able to make more informed travel decisions with up-to-date costs and time considerations.

Detailed Description:

In our application, we implemented a "Plan a Trip" page featuring an Itinerary Generator. This generator accepts user inputs such as the number of days, cities to visit, interests, budget, and number of travellers (adults or children). Using the Google Gemini API, we generate a travel plan based on these inputs. For each attraction suggested by Gemini, a corresponding coordinate is provided, which we use to construct optimized route data displayed on the map beside the travel plan. At the end of the travel plan, users can view the estimated total cost in both USD and TWD (New Taiwan Dollar), giving them a clearer understanding of their overall budget.

Feature 2: AI Bot (Red highlight anything we've changed since M1 and add bracket annotation just after this section to note any changes)

A chatbot powered by the Google Gemini API that provides real-time answers to users' questions. The bot will deliver fast responses in a conversational format to any user's inquiries.

- **Benefits:**

- Users will be able to quickly get answers to commonly asked questions without needing to search through the site manually.
- Users will be able to receive assistance at any time, ensuring timely responses whenever needed.
- Users will be able to save time by receiving answers that are specific to their question and chat history.

Detailed Description:

Our AI bot serves as a helpful assistant that is accessible from every page of the website. It functions primarily as a chatbot, capable of answering user questions as long as they are related to Taiwan. To enhance the user experience, we included a chat history feature that allows the AI to maintain context across the conversation. Users can also reset the chat history at any time to start a new topic. In addition to answering user-typed questions, the AI bot can respond to prompts from two prewritten lists: FAQ and Surprise Me. The FAQ contains a collection of commonly asked questions and their answers. These are handled directly and are not considered

part of the AI's API functionality, so they are excluded from the chat history. The Surprise Me feature presents users with a randomly selected question related to Taiwan. When the user clicks the button, the AI bot provides an answer to the chosen question, offering an engaging way to explore new information.

Feature 3: AI Summary of Attractions

When a user goes to the attraction page, in the description section, they can click a button to retrieve a description of the specific attraction, generating a brief but detailed response.

- **Benefits:**
 - Users will be able to from the attraction page, learn about an attraction's information by clicking a button if they find that the editorial summary from the Google Maps API is not enough.
 - Users will be able to get information more efficiently by seeing concise descriptions that highlight key features.
 - Users will be able to gain a better understanding of each location's cultural or historical context through brief explanations.

Detailed Description:

Since the editorialSummary field fetched from the Google Places API is often very brief and lacks depth, we decided to implement AI-generated descriptions for each attraction. These AI-generated descriptions provide a much more detailed and comprehensive overview of the attractions, offering users richer insights into what makes each location unique. Unlike the editorial summaries, which typically offer a short, surface-level explanation, the AI descriptions include historical background, cultural significance, and contextual details that help travellers better understand the value and experience of visiting each place. This added context not only enhances user engagement but also helps users make more informed decisions when planning their trips. By combining real-time API data with AI-generated content, we aimed to create a more educational and immersive experience for users exploring Taiwan's attractions.

4. CI/CD Pipeline

Provide a detailed description and overview of the CI/CD pipeline. (discuss testing, deployment, and monitoring). How was it used in the project?

For our CI/CD pipeline, we used Vitest for unit and integration testing, Render for continuous deployment, and GitHub Actions to monitor the success of both testing and deployment.

Since we were using React with Vite, we chose Vitest because it is specifically designed to work well with the Vite framework. In our codebase, we ensured that all major components containing data-driven logic had a corresponding `<component_name>.test.tsx` file, organized within dedicated `_tests_` folders. Many of these test files simulated API data fetching and verified that key UI elements were rendered correctly within our page components.

We selected Render as our deployment platform because we wanted to keep our repository private, and it was recommended by our professor. On Render, we configured our environment variables, including API keys, and regularly verified that the main branch was running as expected on the live deployment.

Finally, we used GitHub Actions to ensure automated testing and building were integrated into our development workflow. We added a test job that runs Vitest every time a pull request is made or changes are pushed to the main branch. Additionally, we included a build job to confirm that the web application compiles correctly, catching issues like missing brackets or syntax errors that could cause the build to fail.

5. Testing Strategy

For our testing strategy, we chose Vitest for unit and integration testing because it was specifically developed for the React Vite framework, which our project uses. In the codebase, all major components and relevant files had corresponding `.test.tsx` files located in `_tests_` folders. Most of these test files included test cases that used mock API data along with Vitest's built-in `expect` syntax to assert that the page components functioned correctly. Additionally, we integrated Vitest into our CI/CD pipeline so that GitHub Actions automatically runs the tests on every pull request, ensuring that all tests pass before changes are merged.

6. Project Architecture - DFD Level 1 (Miro)

Miro Link:

https://miro.com/app/board/uXjVLM6JYXI/?share_link_id=135881325611**

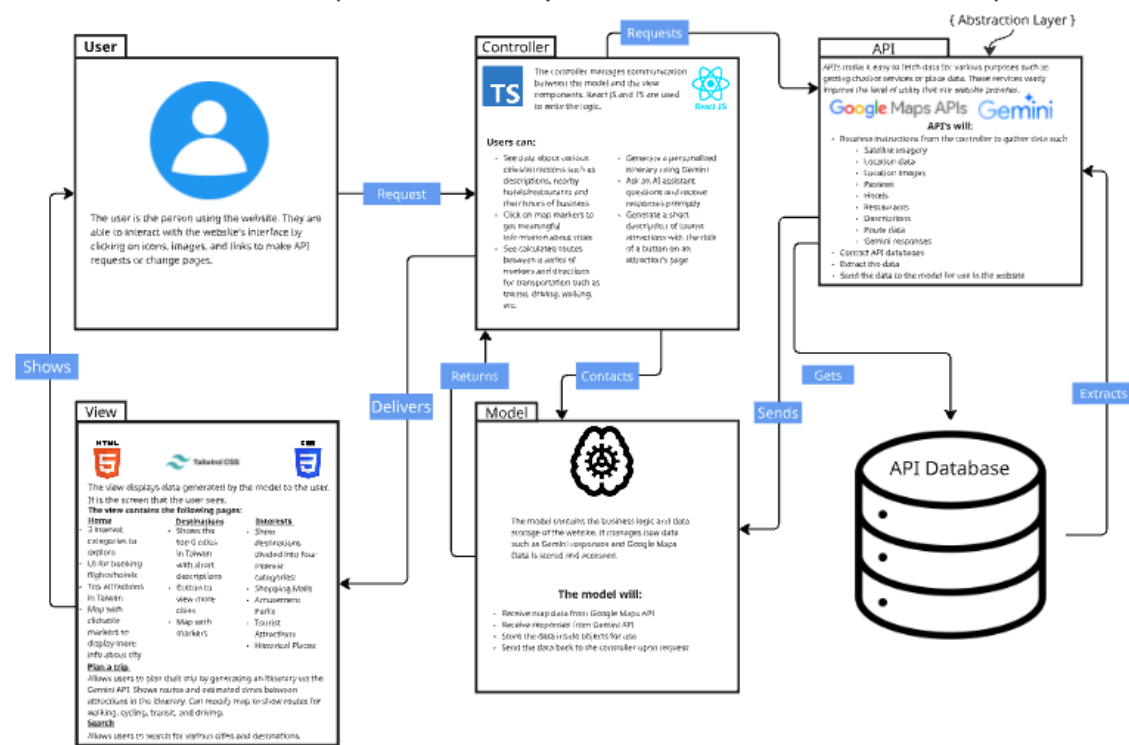
Data Flow Diagram Screenshots:

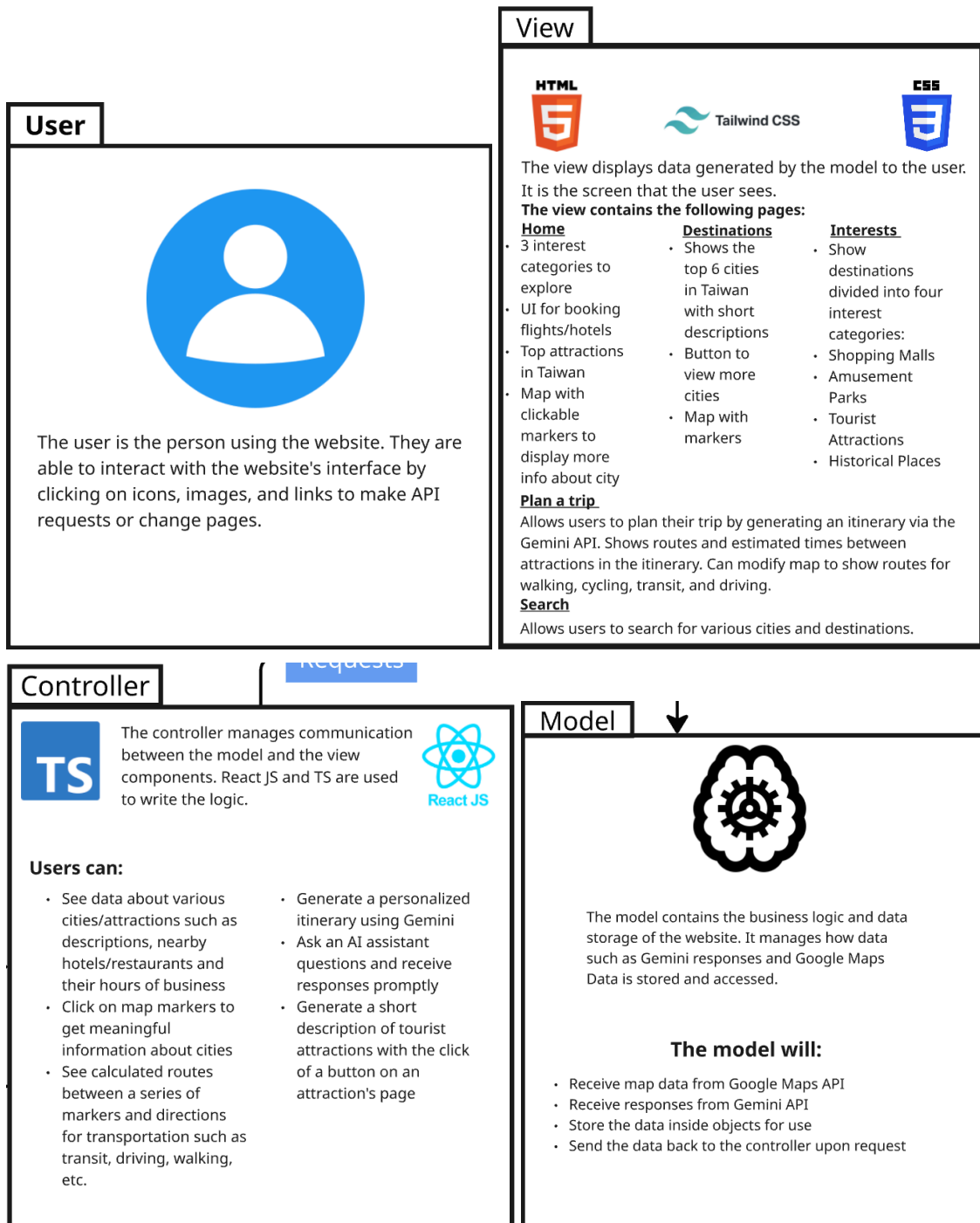
styling to the elements of the pages. When the controller is manipulated, changes to the UI are delivered to the view. The view reflects the current state of internal data structures like strings, arrays, and objects to properly display current data to the user such as chatbot response history, maps with routes, etc.

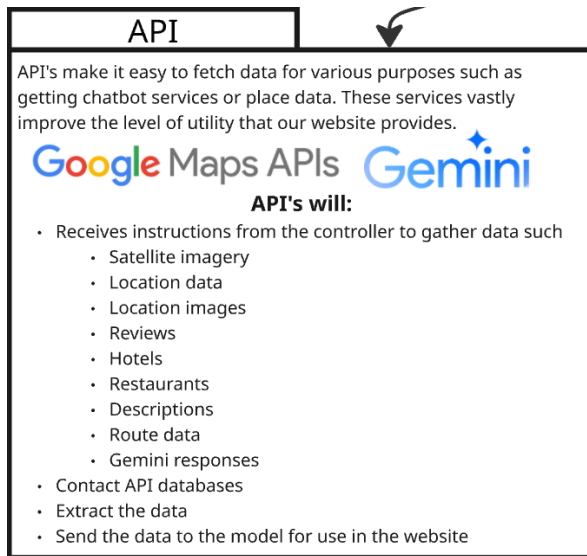
Controller

The controller is manipulated by the user to perform various actions that elicit a visual response on the website. Based on the user's inputs, whether it be text inputs or button clicks, it contacts API's in the model section to execute the appropriate instructions such as fetching a response from the Gemini API or fetching maps and place descriptions from the Google Maps API. The controller's logic was programmed with the React JS, a popular library used for designing UIs which simplified the process of implementing our chosen APIs.

MVC Model Screenshot (Individual Components + Full View are included):







8. List of known bugs and issues (Table)

Bug/Issue	Steps to reproduce	Severity Level	Link to GitHub issue
Itinerary Generator: Generated routes takes a while to load	To reproduce the issue, the user must fill out all the fields in the itinerary generator and press "Generate Itinerary" The route would then take a few seconds to fully load after the trip plan is generated, instead of generating instantly alongside the trip plan	Minor	https://github.com/CMPT-276-SUMMER-2025/final-project-18-flowers/issues/79
Itinerary Generator: Route map not automatically	The issue seems to happen differently depending on the	Minor	https://github.com/CMPT-276-SUMMER-2025/final-project-18-flowers/issues/79

zooming to the generated route markers	device used and has two variants. To reproduce the problem, the user must change the fields in the itinerary generator after previously generating an itinerary in the same session. The new itinerary will then generate new routes, but the map will not move to the updated locations and will instead remain at the previous itinerary's map position (Variant 1 of the issue). In comparison, the other variant of this issue happens when the user must fill out all the fields in the itinerary generator for the first time. In which the itinerary will generate routes, but the map will instead zoom to the center of Taiwan (Variant 2 of the issue).		es/80
Google Maps API Data: Whenever a user right-clicks rather than left-clicks and opens a new tab	1. Find the top bar 2. Navigate to the interests tab 3. Right click on any interest 4. Click "open link in new	Critical	https://github.com/orgs/CMPT-276-SUMMER-2025/projects/3/views/1?filterQuery=known+bug&pane=issue&ite

that relies on Google Places data, any component that depends on that data fails to load in the new tab.	tab" 5. Observe that photos, overview, map, and opening hours are missing		mld=123246320&issue=CMPT-276-SUMMER-2025%7Cfinal-project-18-flowers%7C83
Itinerary Generator: Route map for bicycling shows green lines that represent bike paths which are unintended.	1. Find top bar 2. Navigate to "plan a trip" 3. Fill in all fields in the Itinerary Generator 4. Click "Generate Itinerary" 5. Click on "BICYCLING" on the route map 6. observe numerous green lines appearing on the map.	Trivial	https://github.com/orgs/CMPT-276-SUMMER-2025/projects/3/views/1?filterQuery=bug&pane=issue&itemId=123248998&issue=CMPT-276-SUMMER-2025 final-project-18-flowers 84
Attractions: Taipei City and New Taipei City appear to have the same exact attractions despite being different cities.	1. Find the top bar 2. Navigate to destinations page 3. Click on Taipei City card 4. Make a mental note of attractions you see 5. Go back to destinations page 6. Click on New Taipei City card 7. Notice that both cities have the same attractions	Minor	https://github.com/CMPT-276-SUMMER-2025/final-project-18-flowers/issues/85#issue-3298860725

9. Future Work and Potential Improvements

After the project deadline, what are some future work and potential improvements your group plans to make? This should include any features that weren't implemented and how they could be implemented in the future.

- Improve responsive design for mobile devices
 - At the moment, the responsive design we added to mobile devices is very basic and could benefit from us creating more custom mobile components. Additionally, a lot of the margins and padding could be better adjusted. We'll continue to create .tsx components, use native CSS, and tailwindcss to improve this.
- Empty pages that could benefit from more components
 - A lot of pages, on a computer screen, use a two column page format to display content: one column to display the main content and the other for sub content such as the booking form. However, because we've only made the booking form, the rest of the second column is very empty and could benefit from more components that can make the user experience better. Examples could include a component that navigates the user to other pages of the websites or content such as current events (observe the 5 future work points).
- Improve the destinations map to include more interactivity
 - In the destinations map, we have placed markers corresponding to the destinations, where when clicked displays a card of the destination. The map is however a static map so the map itself is not interactable. To include more interactivity, we plan to make it so when you click on a marker, it zooms in to the corresponding destination, where some markers of its top attractions appear. With this feature, we also plan to add a button to zoom back out. Aside from the map and back to the content card which is closely related to the map, we plan to improve that as well. To improve, we're going to add left and right arrow buttons to the card such that clicking those also changes the highlighted marker in the map to add more interactivity.
- Provide stronger reasons for users to utilize the itinerary generator instead of going to AI tools like ChatGPT
 - The itinerary generator's output itself is lackluster in the sense that our output is not that much more valuable in comparison to asking ChatGPT. We can make our output more valuable by incorporating better styling and constructing a better message to Google Gemini. The

route map which directly relies on the output of the itinerary generator could use colors to differentiate between days. Furthermore, the route map should at least have an option where modes of transportation are dynamically chosen based on distance because it doesn't make sense to display a walking route from one city to another.

- Better input fields for forms
 - At its current state, the itinerary generator's input form could benefit from improved design. Examples include a money slider, more options for different currencies, and better heuristic for empty options on submit.
- More sections that include information about current data and events about Taiwan
 - Currently, our website application doesn't include any information about current data or events about Taiwan. Thus, our website application can be greatly improved by adding it, which can be achieved by researching new APIs which inform users about current events or the weather. Furthermore, by including this, we would have the opportunity to add the current or upcoming Taiwan events to the interest page.
- Improve aesthetic and create a more uniform theme throughout the website
 - Currently, our website application doesn't choose colors from a pre-selected colour palette or theme. Furthermore, some of our CSS colour choices are similar but not the same, resulting in inconsistent design. We believe that by adding a colour palette and theme, our website application would not only look more consistent and visually appealing, but also improve user experience and website memorability.
- Make the route map more accurate and provide more useful information
 - Currently, the route map takes a while to generate on the plan map instead of instantly. Furthermore, there are times when routes are unavailable (No way to get to the location via transit or walking). Thus, basic improvements to this feature would be making routes be able to generate instantly and having correct route data, furthermore allowing different route colors based on which day the route is for (Purple for Day 1 and Orange for Day 2)
- Improve the search bar on mobile devices
 - Due to time constraints and other priorities, on mobile devices, our "search bar" is currently just a button that links the user to the search page. In the future, we plan to create an effect where when you click the search icon, it displays a search bar that allows the user to type in a search query without hindering the mobile experience.
- Add animation to interests page when clicking "Home -> Interests"

- When navigating to the interests page, there is no animation to help make the website feel more professional. We plan to add a fade in animation later down the road to improve the look and feel of the website.
- Add images to search results
 - In the search page, the search results do not have images related to the result. The plan is to have every single search result on the page have a corresponding image to help users visualize the attraction.
- Better error prevention for booking flights
 - Right now, when a user searches for a flight on our website, they can set the return date to be a time before the departure date, which obviously doesn't make any sense. This causes the flight search results on booking.com to be completely empty. Most users would not choose a return date that is earlier than the departure date. But just in case this does happen, the book a flight panel should be improved by preventing the user from choosing a return date that is earlier than the departure date.
- Improve visibility of system status for input fields in itinerary generator form
 - When the user wants to generate an itinerary, they must fill in all fields in the form to do so. If they try to generate one without accomplishing this task, the system prevents the action and displays a red error message in the output area. Though unlikely, the user might be confused as to which fields are missing, and thus, become frustrated when their itinerary is failing to generate. This feature could be further improved by displaying a red error message inside or around every field that is unfilled to make it clear to the user which fields are missing.

10. Lessons Learned and Takeaways

What are some challenges the group/individuals faced during the project? How did you overcome them?

- Communication Challenges
 - During M1 and M2, especially during the implementation phase, our group faced numerous challenges regarding communication. One aspect of communication we struggled with was the clarity of who is responsible for what task. This would not only delay our progress but also cause frustration to those who do more work as a result, and those who don't understand what exactly they need to do. Another aspect we struggled with was the lack of transparency, whether that relates to a major change made or if a section of the code was generated by AI.

This makes collaborating difficult, as one person might expect something to be in a different state when it was already modified. When asking a question about a part of the code, a person doesn't fully understand it because it was assisted marginally by AI.

- To solve the first aspect, usually one of the group members would point out their frustration and then everyone would discuss the issues and what needs to happen. In this case, what's going wrong is that some people don't know what they are tasked with. The project manager, alongside the group, will specify exactly what they should do. In the second aspect, we fixed the problem by ensuring that whenever a major commit or a new branch is made, the group chat is notified of this. Additionally, to solve the lack of understanding of certain sections of code, we took the time to sit down and write detailed comments throughout our codebase.
- Redundant, In-person Meetings
 - For every week of this semester, our groups generally had 2 in-person meetings every week, specifically on Wednesday and Friday. However, we often found that our meetings were hindered by distractions and lacked meaning, especially for members who had to commute to the Surrey campus solely for the Friday meeting. We believe the reason this occurred is that no good, structured plan was prepared before the meeting.
 - To combat this, we made the in-person meetings hybrid, meaning members can either choose to get on a call or come to the usual in-person room we always book at campus. Additionally, we decided to make the meeting more concise rather than enforcing the duration to be 1 hour, to avoid wasting anyone's time. We also started utilizing scrum meetings to clarify what everyone has done, is working on, and anything they're struggling with.
- Scope Creep
 - At the start of the project, as a group, we decided to do a travel guide for Taiwan with features from the Google Maps API and Gemini API; however, as the semester progressed, we quickly realized how much work we had given ourselves with the scope. This became extremely apparent during the peer testing session, when we saw the other group's projects
 - To solve this challenge, we didn't implement features/sections of the application that weren't as important as our core features (e.g., itinerary generator, AI chatbot, etc.). Some of the removed features/sections include the Plan a Trip section on the home page, the money slider for the itinerary generator, and extensive options in our booking component.

- Lack of Testing Experience
 - During our project, we decided to use Vitest after the M1 report; however, we quickly realized that none of us had any prior experience or knowledge of Vitest. Furthermore, even after spending time trying to understand Vitest, we still made no progress. This challenge was only further amplified when some of our members had no prior experience with HTML/JavaScript/CSS/React. Additionally, all of our group members had zero experience working with APIs.
 - Thus, ultimately, to not waste more time on this challenge, we decided to use ChatGPT to generate mock data tests for our API features and help us overcome this challenge.
- Managing APIs and Credits Usage
 - At the start of the project's code phase, we didn't prioritize using mock data or cache to save free API credits, which caused us to waste precious credits in testing the API data. This became a serious problem when the original group member's free credits were instantly used up from an API key leak while testing the Gemini API.
 - Thus, after we let another member create new API keys for more free credits, we immediately thought of ways to save credits and prevent this problem from occurring again, with the solution being the usage of caches for when the data of a location will be the same for each session and mock data to test new code added to an API feature.

11. Appendix

1. **Provide a list of all group members and their detailed contributions to the project**

Alex Lee

Contributions to the Task:

- Milestone 0: Set up the project repository on GitHub, which included adding docs, src, and misc folders to the repository. Furthermore, wrote the cover page and created the initial layout for the Project Proposal Report, including the project title, group member names, and GitHub repository link.
- Milestone 1: Finalized the 2 APIs and identified three features for each API, ensuring that each of the features is within the scope of the project by including three benefits for each feature. Transformed the low-fidelity storyboard into a mid-fidelity prototype using Figma. Furthermore, completed the risk assessment by identifying 15 unique risks categorized by severity as low, medium, and high.
- Milestone 2:

- Worked on the Implementation of API 1 Features, which include Feature 2 (Interactive Map) and Feature 1 (Integrating Google's Vast Database on Places to the application). Added GitHub Actions to the application's workflow to aid the CI/CD pipeline alongside Ryan. Furthermore, worked on the search page, booking components and added components to both the home page and interests page. Lastly, worked on the overall UI/CSS layout and responsive design for all pages.
- Edited the video of M2, worked on the CI/CD, Lessons Learned and Takeaways, and Future Work and Potential Improvements of the M2 Report

Ikkyu Li

Contributions to the Task

- Milestone 0: Created detailed user personas to represent different types of travellers, including a student, elderly, explorer, and hiker, by outlining their needs, tech experience, and pain points. Furthermore, she wrote the corresponding user stories to capture their goals and expectations when using the application.
- Milestone 1: Constructed the Work Breakdown Structure (WBS) to visualize the tasks and responsibilities across milestones. Furthermore, worked on and completed both level 0 and level 1 of the Data Flow Diagrams (DFDs)
- Milestone 2:
 - Tested and reviewed the code of other members, worked on the Interest and Attractions page. Furthermore, worked on the API 2 Features, which include AI Description (Feature 3), AI Chatbot (Feature 2) and bug fixes for the Itinerary Generator (Feature 1). Additionally, on the CSS for the Attractions page and Chatbot Feature. Lastly, added comments to the main files of the application
 - Worked on the DFD and forms survey part of the appendix of the M2 report, as well as the subtitles for the M2 Video

Ryan Fu

Contributions to the Task

- Milestone 0: In charge of the submission of Milestone 0, which involved reviewing all the completed tasks, checking that each required file was present and properly formatted, and making sure everything met the project requirements, which ensured that all files were complete and submitted on time.
- Milestone 1: Created and completed the project schedule with start/end dates and due dates for all tasks based on the WBS milestone breakdown. Assigned

and accomplished the Appendix With Additional Information, which included a changelog and group contributions. Furthermore, handled the final submission of the planning report for Milestone 1, while ensuring that all files were correctly submitted.

- Milestone 2:
 - Worked on the Implementation of API 1 Features, which include Feature 2 (Interactive Map) and Feature 3 (Return Route Data). Added GitHub Actions to the application's workflow to aid the CI/CD pipeline alongside Alex. Furthermore, added a filter to the attractions so that Google Maps API only provides results in English. Also implemented Feature 3 of API 1 (Return Route Data) into Feature 1 of API 2 (Itinerary Generator) and made bug fixes on the Plan a Trip page and the cache. Lastly, added vitest files to all API related files of the project
 - Worked on the Appendix, Lessons Learned and Takeaways, and Future Work and Potential Improvements of the M2 Report

Tian Zheng

Contributions to the Task

- Milestone 0: Wrote the explanation for the tech stack we chose as a group which included HTML, CSS, Tailwind, JavaScript, and React. Furthermore, the explanations described how each tool will be used in the project, focusing on things like making the site responsive and organized.
- Milestone 1: Researched and selected the Agile Kanban SDLC model as the final project's development approach, which includes a detailed explanation of why it was a good fit, highlighting its flexibility, teamwork, and how it helps track tasks using GitHub. Furthermore, worked on and completed the MVC model diagram to illustrate how the project is organized and how data flows between its various components, including the user interface, logic, and database.
- Milestone 2:
 - Worked on the implementation of API 2 Feature 1 (Itinerary Generator), along with bug fixes and CSS styling for API 2 Feature 2 (AI Chatbot). Additionally, addressed and fixed bugs related to API 2 Feature 3 (AI Description). Lastly, deleted all unusable code and console logs of the code for the final deployment product
 - Worked on the SDLC, MVC, Future Work and Potential Improvements, and known bugs and issues of the M2 Report

Group Contributions

Contributions to the Task as an entire group

- Milestone 0: Worked together to answer the questions and sign the group contract, research and describe the APIs and their possible features (with each member being assigned on researching and describing one API), decided on the front-end tech stack to use for the project as a group, and create the low-fidelity storyboard using Figma.
- Milestone 1: Worked together and checked in with each other through weekly meetings. Additionally, each member supported others by giving reviews and feedback on their tasks to ensure everything was complete and free of issues.
- Milestone 2: Worked together on the video script and recording

2. Provide a changelog table that includes any revisions since the proposal

Date	Revisions made	Reason for revision
6/27/25	Added TypeScript as a tech stack that will be used for the final project	Adding TypeScript to the tech stack improves reliability, team collaboration, and long-term maintainability. Furthermore, TypeScript allows us to declare data types for variables we'll use for user input data, which aids us in detecting errors easily. These benefits will greatly support the progress and the success of this project/application.
7/1/25	Replaced the Street View feature in Google Maps API with a Route Calculation API	The original Google Maps API feature was replaced because the new feature is much more useful for helping users plan their trips and navigate around Taiwan.
7/1/25	Merged the Custom Budget Trip feature into the Itinerary Generator feature for Google Gemini API	These two features were merged into one because they had significant overlap between them. By combining them into one feature, it will simplify the

		user experience and help reduce the project scope for the team.
7/1/25	Added AI Summary of Places as a new Google Gemini API feature	Gave users quick summaries of locations to help users with research and decision-making.
7/1/25	Rewrote and renamed the AI Helper/FAQ feature using Google Gemini API	Improved clarity and alignment with the actual functionality of the feature.
7/2/25	Refined the scope of the wireframe during the transition from low-fidelity storyboard to mid-fidelity prototype	To focus on essential pages and reduce visual clutter. Some features were simplified or removed to keep the prototype and application realistic, usable, and achievable within the project time limit.
7/3/25	Clarified and assigned group roles and responsibilities for Milestone 1 and all milestones onwards	Ensured accurate documentation of each member's work for submission and ensured every member had an equal amount of work.
7/16/25	Changed and was specific on Feature 2 of API 1 (Interactive Dynamic Maps)	Our original Feature 2 API 1 didn't mention static maps, although we used them extensively. Furthermore, we got rid of the marker hover feature in place of content cards. Lastly, the filter/customization of the static map was completely removed since it didn't really fit what we were aiming for.
7/23/25	Slightly adjusted one of the benefits of Feature 2	Specifically, the user can receive answers not only

	API 2 (AI Chatbot)	from their questions but also from their chat history.
7/26/25	Changed Feature 3 API 2 (AI Summary) to be more beneficial to users	We decided that instead of showing an AI summary when you hover over a marker, we moved the AI Summary to the attractions page. Now users can click a button to get an AI summary of the attraction.

3. Peer testing feedback received from classmates (survey results)

Feedback form: <https://forms.gle/QARt2NNDbdURGKVE8>

- What group are you part of?
- How was your overall experience using the website?
- How user-friendly did you find the interface? 1-5 Scale: Very difficult to use ↔ Very easy to use
- Did you encounter any bugs or issues? If yes, please describe what happened and the steps taken to encounter the issue.
- What did you like the most about the website?
- What did you dislike or find frustrating?
- Are there any navigation, design, or clarity improvements that you might suggest?
- If any, what features would you suggest improving or adding?
- Any final suggestions or comments?

Google Sheets Link of Feedback Forms:

<https://docs.google.com/spreadsheets/d/1OUBKvvQx4JCqspHyYa1qN2wJUmxdsVBDeKh8ApB2f4/edit?usp=sharing>

What group are you part of?

4 responses

5

17

3

17-Sunsets

How was your overall experience using the website?

4 responses

Good

Overall smooth experience, not much difficulty finding where I needed to go.

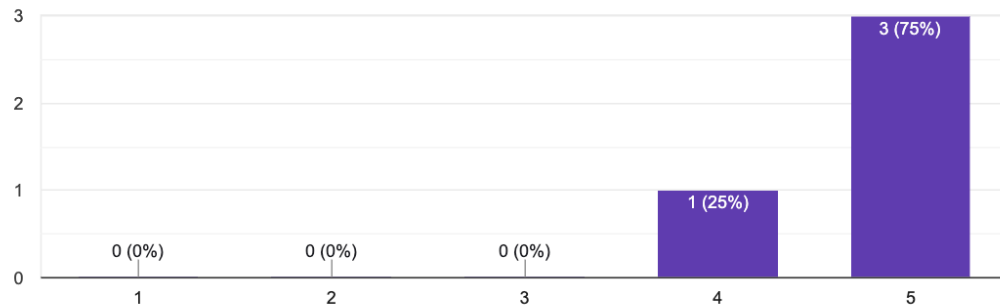
Good experience

Perfect!

How user-friendly did you find the interface?

 Copy chart

4 responses



Did you encounter any bugs or issues?

If yes, please describe what happened and the steps taken to encounter the issue.

4 responses

- 1: For book a flight: when i tried to type in the year, month, and day in 2025-08-05, it bugged out and year became 202508 and no more entries.
- 2: Display issue for: generate your itinerary, it says 3 days: days trip. I think it would be better to say simply 3 days trip.
- 3: Display currency exchange: generate your itinerary: for USD to the taiwan currency might be better.

N/A

No major bugs, just map to resteraunt not opening right

None at all

What did you like the most about the website?

4 responses

Very nice display and overall UI. Well designed.

I liked the number of options available to me for traveling and how the home page is organized so I can easily find those different options.

Overall UI and convenience

The design theme reflect Taiwan culture while not being too Information dense

What did you dislike or find frustrating?

4 responses

The text were overall small and a little tight. If possible make bigger and more concise.

N/A

The font on some pages, minor css.

The ai chat bot responses could be improved, but just the appearance. Maybe use a bubble like container for styling?

Are there any navigation, design, or clarity improvements that you might suggest?

4 responses

More concise text and bigger font.

It wasn't immediately clear if I could interact with the map at all other than switching between modes of transportation.

Add button for zoom in/out

Nope, everything was easy to navigate.

If any, what features would you suggest improving or adding?

4 responses

A saving function so i could save the days suggest to my phone or somewhere to view again

Regarding the map, I think it would be nice if I could zoom in and see the routes better, and some UI to indicate that is an option.

Add button for zoom in/out.

Examples of pre generated itineraries.

Any final suggestions or comments?

3 responses

Spoke in person.

Very well done, it looks like professional website and works smoothly

Awesome project!

4. Any additional diagrams, charts, or tables that were not included in the main report

N/A